Uniform two-sided resonance-kernel integral bounds in dimensions at least three

Marnie Smith

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Editorial note. Only the two original integral inequalities(one),(two) in the Control of resonances lemma are counted. The direct Schur operator corollary1571 is omitted from this counted unit rather than promoted to another task.

Editorial note. Source1564 explicitly credits analogous estimates in Bedrossian-Masmoudi-Mouhot2018, Lemmas3.1 and3.2, and describes this proof as the extension to d>=3. Preserve that mathematical attribution and compare the exact generalized scope before uniqueness admission.

Editorial note. The symbol x introduced at1584 denotes the auxiliary exponent1/(1-zeta), not a spatial integration variable. The implicit constants may depend on fixed d,m,beta and the chosen exponents, but not on T-star,t,s,k,ell.

Editorial note. The changing-time integrals near1597/1627 use ordinary improper-integral estimates. No generated expansion of those estimates is being presented as author text.

Editorial scope. One counted pair of Schur integral estimates. The immediately stated operator corollary1571, full nonlinear Hartree theorem and physical-space decay corollary are not separately counted.

Allowed standard prerequisites (editorial).

- Elementary weighted convolution estimates. Split Euclidean integrals into a neighborhood of each center and their complement; power tails are integrable when their combined exponent exceeds the dimension. Singular factors|x|^(-1/2) are locally integrable in the stated dimensions.. Check the one-dimensional parallel integrals and full-dimensional nonresonant integrals after the author changes of variables. No time-response kernel bound is supplied.
- Tonelli, Cauchy-Schwarz and linear changes of variables. Integrate the positive kernel and rescale time along a nonzero parallel frequency; zero frequencies are handled separately.

0.1 Quantum operators and norms

To formulate phase-mixing and scattering estimates in the quantum setting, quantum analogues of classical derivatives and weighted Sobolev norms are introduced. The standard and double Japanese brackets are defined by $\langle x \rangle :=$

$(1+|x|^2)^{1/2}$ and $\langle x,y\rangle := (1+|x|^2+|y|^2)^{1/2}$, respectively. For a multi-index $\alpha=(\alpha_1,\dots,\alpha_d)\in\mathbb{N}^d$, write $|\alpha|=\alpha_1+\cdots+\alpha_d$ and $D^\alpha=\partial_1^{\alpha_1}\cdots\partial_d^{\alpha_d}$.

Throughout, integrals such as $\int_x dx$ denote integration over $\mathbb{R}^d$ in the x variable. The following conventions are adopted for the Fourier transform of functions $g=g(v)$ and $f=f(x,v)$:

$$\widehat{g}(p) := \int_v e^{-ip\cdot v} g(v)\,dv, \qquad \widehat{f}(k,p) := \int_x\int_v e^{-ik\cdot x}e^{-ip\cdot v} f(x,v)\,dvdx.$$

Lemma 0.1 (Control of resonances)**.** *Fix dimension $d\geq 3$ and let $T^*\geq 0$, $m\geq \max\{1,(d-3/2)/2\}$ and $\beta\geq d+6$. The following integral inequalities hold:*

$$\sup_{t\in[0,T^*],k\in\mathbb{R}^d}\int_0^t\int_{\ell\in\mathbb{R}^d}\frac{\langle s\rangle|k|^{1/2}|\ell|^{1/2}}{\langle\ell\rangle^m\langle k-\ell,kt-\ell s\rangle^\beta}d\ell ds\lesssim 1 \tag{1}$$

and

$$\sup_{s\in[0,T^*],\ell\in\mathbb{R}^d}\int_s^{T^*}\int_{k\in\mathbb{R}^d}\frac{\langle s\rangle|k|^{1/2}|\ell|^{1/2}}{\langle\ell\rangle^m\langle k-\ell,kt-\ell s\rangle^\beta}dkdt\lesssim 1. \tag{2}$$

Proof. To establish inequality (1), fix $t\in[0,T^*]$ and $k\in\mathbb{R}^d$. If $k=0$, the integral vanishes, so assume $k\neq 0$. The integral over early times is controlled as follows:

$$\int_0^1\int_\ell\frac{\langle s\rangle|k|^{1/2}|\ell|^{1/2}}{\langle\ell\rangle^m\langle k-\ell,kt-\ell s\rangle^\beta}d\ell ds\lesssim\int_\ell\frac{|k|^{1/2}|\ell|^{1/2}}{\langle\ell\rangle^m\langle k-\ell\rangle^\beta}d\ell\lesssim\int_\ell\frac{|\ell|^{1/2}}{\langle\ell\rangle^{m-1/2}\langle k-\ell\rangle^{\beta-1/2}}d\ell\lesssim 1,$$

where $\langle k\rangle\lesssim\langle\ell\rangle\langle k-\ell\rangle$ was used.

Assume $T^*>1$. Decompose $\ell\in\mathbb{R}^d$ into parallel and orthogonal components relative to k as $\ell_\parallel := (k\cdot\ell)k/|k|^2$ and $\ell_\perp := \ell-\ell_\parallel$. Define $I_R\subset\mathbb{R}^d$ as a cylinder around the line spanned by k with radius shrinking over time, and set $I_{NR}=I_R^c$, where

$$I_R=\big\{\ell:|\ell_\perp|<{}^{|k|^b}\!/_{\langle s\rangle^\zeta}\big\}\quad\text{and}\quad I_{NR}=\big\{\ell:|\ell_\perp|\geq{}^{|k|^b}\!/_{\langle s\rangle^\zeta}\big\},$$

with $b:=(1-\zeta)/2$, where ζ is chosen to satisfy

$$\frac{2}{d-1/2}<\zeta<1\quad\text{and}\quad x:=\frac{1}{1-\zeta}<\frac{11}{2}.$$

Such a choice is possible since $(1-\zeta)^{-1}\to(2d-1)/(2d-5)\leq 5$ as $\zeta\downarrow 2/(d-1/2)$, for all $d\geq 3$.

The integral over the cylinder I_R satisfies, using $|\ell|^{1/2}\lesssim|\ell_\parallel|^{1/2}+|\ell_\perp|^{1/2}$,

$$\begin{aligned}\mathscr{I}_R &:= \int_1^t\int_{I_R}\frac{\langle s\rangle|k|^{1/2}|\ell|^{1/2}}{\langle\ell\rangle^m\langle k-\ell,kt-\ell s\rangle^\beta}\,d\ell ds\\ &\lesssim\int_1^t\int_{I_R}\frac{\langle s\rangle|k|^{1/2}}{\langle\ell_\parallel\rangle^m\langle k-\ell_\parallel,kt-\ell_\parallel s\rangle^\beta}\big(|\ell_\parallel|^{1/2}+|\ell_\perp|^{1/2}\big)\,d\ell ds.\end{aligned}$$

Applying the bound for $|\ell_\perp|$ on I_R and the volume of its cross-section gives

$$\mathscr{I}_R\lesssim\int_1^t\int_{-\infty}^{\infty}\frac{\langle s\rangle|k|^{1/2}}{\langle\ell_\parallel\rangle^m\langle k-\ell_\parallel,kt-\ell_\parallel s\rangle^\beta}\frac{|k|^{(d-1)b}}{\langle s\rangle^{(d-1)\zeta}}\Big(|\ell_\parallel|^{1/2}+\frac{|k|^{b/2}}{\langle s\rangle^{\zeta/2}}\Big)\,d\ell_\parallel ds,$$

leading to two contributions $\mathscr{I}_1$ and $\mathscr{I}_2$. For $\mathscr{I}_1$, the change of variable $s \mapsto s/|\ell_\parallel|$ and $(d-1)\zeta \geq 1$ yield convergence under the conditions $\beta+m-2(d-1)b > 3/2$ and $\beta, m \geq 1/2+(d-1)b$. The second term $\mathscr{I}_2$ converges due to the choice of ζ, ensuring that $1/\langle s\rangle^{(d-1/2)\zeta-1}$ is integrable, for $\beta+m > 2+2(d-1/2)b$ and $m, \beta \geq 1/2+(d-1/2)b$.

On I_{NR}, for $s \geq 1$, it holds that $|kt-\ell s| \gtrsim s|\ell_\perp| \gtrsim |k|^b\langle s\rangle^{1-\zeta}$, so

$$\begin{aligned}\mathscr{I}_{NR} &:= \int_1^t \int_{I_{NR}} \frac{\langle s\rangle |k|^{1/2}|\ell|^{1/2}}{\langle \ell\rangle^m \langle k-\ell, kt-\ell s\rangle^\beta}\, d\ell ds \\ &\lesssim \int_1^t \int_{I_{NR}} \frac{\langle s\rangle |k|^{1/2}|\ell|^{1/2}}{\langle \ell\rangle^m \langle k-\ell, kt-\ell s\rangle^{\beta-1/(2b)} |k|^{1/2} \langle s\rangle^{(1-\zeta)/(2b)}}\, d\ell ds.\end{aligned}$$

Using $(1-\zeta)/(2b)-1=0$, and applying the change of variable $s \mapsto s/|\ell|$, the integral $\mathscr{I}_{NR}$ converges for $\beta+m-1/(2b)-1/2 > d$.

To establish inequality (2), it is observed that:

$$\begin{aligned}&\sup_{s\in[0,T^*],\ell\in\mathbb{R}^d} \int_s^{T^*} \int_{k\in\mathbb{R}^d} \frac{\langle s\rangle |k|^{1/2}|\ell|^{1/2}}{\langle \ell\rangle^m \langle k-\ell, kt-\ell s\rangle^\beta}\, dkdt \\ &\qquad\qquad \leq \sup_{s\in[0,T^*],\ell\in\mathbb{R}^d} \int_0^{T^*} \int_{k\in\mathbb{R}^d} \frac{\langle t\rangle |k|^{1/2}|\ell|^{1/2}}{\langle \ell\rangle^m \langle k-\ell, kt-\ell s\rangle^\beta}\, dkdt.\end{aligned}$$

For fixed $s \in [0,T^*]$ and $\ell \in \mathbb{R}^d$, the integral over early times is estimated as

$$\int_0^1 \int_k \frac{\langle t\rangle |k|^{1/2}|\ell|^{1/2}}{\langle \ell\rangle^m \langle k-\ell, kt-\ell s\rangle^\beta} dkdt \lesssim \int_k \frac{|k|^{1/2}|\ell|^{1/2}}{\langle \ell\rangle^m \langle k-\ell\rangle^\beta}\, dk \lesssim \int_k \frac{|k|^{1/2}}{\langle k\rangle^\beta}\, dk \lesssim 1.$$

Assume $T^* > 1$. If $\ell = 0$, the integral vanishes, so assume $\ell \neq 0$. Decompose $k \in \mathbb{R}^d$ into parallel and orthogonal components relative to ℓ, defining $k_\parallel := (\ell\cdot k)\ell/|\ell|^2$ and $k_\perp := k - k_\parallel$, and partition

$$J_R = \big\{k : |k_\perp| < {}^{|\ell|^b}\!/_{\langle t\rangle^\zeta}\big\} \quad \text{and} \quad J_{NR} = \big\{k : |k_\perp| \geq {}^{|\ell|^b}\!/_{\langle t\rangle^\zeta}\big\},$$

with ζ, b and x as in the proof of (1).

The integral over J_R satisfies

$$\begin{aligned}\mathscr{J}_R &:= \int_1^{T^*} \int_{J_R} \frac{\langle t\rangle |\ell|^{1/2}}{\langle \ell\rangle^m \langle k_\parallel-\ell, k_\parallel t-\ell s\rangle^\beta} \left(|k_\parallel|^{1/2} + |k_\perp|^{1/2}\right) dkdt \\ &\lesssim \int_1^{T^*} \int_{\mathbb{R}} \frac{|\ell|^{1/2}}{\langle \ell\rangle^m \langle k_\parallel-\ell, k_\parallel t-\ell s\rangle^\beta} \frac{|\ell|^{(d-1)b}}{\langle t\rangle^{(d-1)\zeta-1}} \left(|k_\parallel|^{1/2} + \frac{|\ell|^{b/2}}{\langle t\rangle^{\zeta/2}}\right) dk_\parallel dt,\end{aligned}$$

leading to two contributions, $\mathscr{J}_1$ and $\mathscr{J}_2$. Following the change of variable $t \mapsto t/|k_\parallel|$ in $\mathscr{J}_1$, the integrals converge under the conditions $m \geq (d-1/2)b+1/2$ and $(d-1/2)\zeta-1 > 1$.

On J_{NR}, for $t \geq 1$, it holds that $|kt-\ell s| \gtrsim t|k_\perp| \gtrsim |\ell|^b\langle t\rangle^{1-\zeta}$, so

$$\begin{aligned}\mathscr{J}_{NR} &:= \int_1^{T^*} \int_{J_{NR}} \frac{\langle t\rangle |\ell|^{1/2}|k|^{1/2}}{\langle \ell\rangle^m \langle k-\ell, kt-\ell s\rangle^\beta}\, dkdt \\ &\lesssim \int_1^{T^*} \int_{J_{NR}} \frac{|\ell|^{1/2}|k|^{1/2}}{\langle \ell\rangle^m \langle k-\ell, kt-\ell s\rangle^{\beta-x} |\ell|^{bx} \langle t\rangle^{x(1-\zeta)-1}}\, dkdt.\end{aligned}$$

Since $x(1-\zeta) = 1$ and $bx = 1/2$, the change of variable $t \mapsto t/|k|$ shows that the integral is uniformly bounded provided $\beta - x - 1/2 > d$. By the choice $x < 11/2$, this condition follows from $\beta \geq d + 6$. The remaining conditions obtained above follow from $\beta \geq d + 6$ and $m \geq \max\{1, (d - 3/2)/2\}$. □

References

[1] J. Bedrossian, N. Masmoudi & C. Mouhot - "Landau damping in finite regularity for unconfined systems with screened interactions", Comm. Pure Appl. Math. 71 (2018) no. 3, p. 537-576